Student Performance Predictor

1. Introduction:-

In the modern education system, predicting student performance has become an important tool for teachers, parents, and educational institutions. Accurate predictions can help identify students who may need additional support and optimize their learning. This project aims to develop a simple predictive system that can tell whether a student will **pass or fail** based on the amount of time they spend on their academics (study hours).

The **Student Performance Predictor** is built using the **Python programming language** and implements **linear regression** to model student outcomes. The project uses fundamental concepts in AI/ML, including data preprocessing, model training, and prediction.

2. Objective:-

The main objectives of the project are:

- Predict whether a student will pass or fail based on historical performance data.
- Apply linear regression, to a real-world dataset.
- Learn how to prepare, train, and evaluate a predictive model.

3. Tools and Technologies Used:-

1. Python 3.14 – main programming language.
2. Spyder IDE – for coding and testing.
3. CSV (Comma-Separated Values) – to store student data.
4. Linear Regression – machine learning model to predict outcomes.
5. Git and GitHub – version control and repository management.

4. Project Structure:-

The project has the following structure:-

```
Student Performance Predictor/
├── Source code/
│   ├── main.py          │
└── model.py         ├──
data/
│   └── students.csv
├── README.md
└── Project_Report.pdf
```

5. Data set

My data set includes `data.csv:-`

Feature name	Description
Study hours	No of hours a student studies
Attendance	Attendance percentage
Marks	Marks scored in tests

Example rows:-

Study hours	Attendance	Marks
2	60	45
3	65	50
4	70	55
5	75	60
6	80	65

Note:-

The Pass/Fail label is derived from marks

(e.g., marks >= 50 = Pass, marks < 50 = Fail).

6. Methodology:-

1. Load the data from the CSV file.
2. Preprocess the data (add a Pass/Fail column based on marks).
3. Train a linear regression model using `study_hours`, `attendance`, and `marks` as input features.
4. Predict whether a student will pass or fail.
5. Evaluate the model's performance.

7. How to Run:-

Step 1:- Clone the repository by opening command prompt

```
git clone
https://github.com/Pushkar10105/Student-
performance-predictor.git
```

Step 2:- Navigate to the source folder

```
cd Student-performance-predictor/"Source code"
```

Step 3:- Run the program

```
python main.py
```

8. Screen shots :-

First when we run the program we get a menu interface asking us to input the hours studied by a student

```
===== Student Performance Predictor =====

Enter number of study hours:
```

Then now we can enter the requested data and see what our model predicts

```
===== Student Performance Predictor =====

Enter number of study hours: 10

Predicted Marks: 85.0
Result: ✅ PASS

========================================
```

This is our final working of our model

9. Conclusion

This project demonstrates how linear regression can help predict student pass/fail outcomes. It provides hands-on experience with Python, CSV handling, and basic machine learning concepts, while allowing practical application of study data.

Developing this project was a very fun and immersive practical learning experience i got to use linear regression in this project

Finally this project was Developed by:-

SAGNIK ADHIKARI

Registration no:- 25BAI11387